

Financial Econometrics

Writing tips

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Overview

- ▶ Disclaimer
- ▶ Big picture
- ▶ Organization
- ▶ Writing
- ▶ Advice for empirical work

Disclaimer

- ▶ Opinion versus science
 - ▶ Personal experience, following presented advice leads to improvement for master theses
- ▶ I am not a native speaker. Your reader might not be a native speaker
- ▶ Why care?
 - ▶ Exam
 - ▶ Master thesis (perspective of this lecture)
 - ▶ Any writing

Big picture

General

- ▶ One central and novel contribution/finding
 - ▶ Can you write it down in a paragraph?
 - ▶ Realization: How much of what I did has to be thrown out?

General

- ▶ Your readers are busy and impatient
 - ▶ How likely is it that they sit down and read everything from beginning to end, the first time they sit down with your paper?
 - ▶ Readers skim $\Rightarrow$ make it easy to skim
 - ▶ Readers want to find the main result fast $\Rightarrow$ help them

General

1. “Joke” or “novel” style:
 - ▶ Long windup
 - ▶ Punchline in the end
 2. “Triangular” or “newspaper” style
 - ▶ Most important part in the beginning
 - ▶ Details later
- ▶ Your work should follow style 2
 - ▶ Put the punchline right up front
 - ▶ Slowly explain the joke

General

- ▶ A paper is not a travelogue of your work!
 - ▶ Is it important what you tried that did not work?

General

- ▶ First drafts of exams/assignments/theses/papers usually include all the aforementioned pitfalls
- ▶ Internal resistance to some of the advice here

A good empirical Master thesis (from my perspective)

- ▶ A clear, narrow, and interesting research question
 - ▶ Guides decision making on data, model, writing, etc.
 - ▶ No results are interesting
- ▶ One main finding/contribution
- ▶ Page count does not matter, content is the decisive factor
 - ▶ Is it better to make the identical point on X pages or on $2 \cdot X$ pages?

A great Master thesis

- ▶ Good identification: Can you say anything causal versus correlation?
- ▶ Mechanism: Evidence of the economic mechanism/channel that leads to the main finding

Example of a great Master thesis - Abstract

According to Norwegian law, it is illegal to strategically underprice (in Norwegian “Lokkeprise”) real estate, which involves setting an asking price that the seller is not willing to accept and/or a low asking price compared to the expected market value of a dwelling. We analyse the 2016 policy shift of abolishing third party surveyor valuations (in Norwegian “Verditakst”) on underpricing in Oslo and Bergen. We argue that real estate agents prefer a quicker sale compared to sellers, and are therefore interested in setting a low asking price to attract more bidders. Further, we suggest that abolishing surveyor valuations increases information asymmetry between the agent and the seller, thus enhancing the agent’s opportunities to underprice. Using both non-experimental and quasi-experimental analyses, we find strong evidence in favour of increased underpricing from removing the surveyor valuation in Oslo. The average increase in the spread between the sales price and asking price is estimated at 2 - 3 percentage points, corresponding to NOK 100 000 - 150 000.

Example

- ▶ RQ: Does the removal of surveyor valuation lead to more underpricing?
- ▶ Mechanism
 - ▶ Brokers prefer fast sales, and hence like to underprice
 - ▶ Removing third party valuation increases information asymmetry, and
 - ▶ Hence brokers have more opportunity to underprice
- ▶ Empirical set-up
 - ▶ Policy shift in 2016
 - ▶ Difference-in-Difference approach comparing prices in the month before and after the policy change
 - ▶ Treatment: Oslo, Control group: Other city
 - ▶ Result: Underpricing increases by 2-3 percentage points
- ▶ The abstract is a bit long but well written

Organization

Organization

- ▶ Abstract
- ▶ Introduction
- ▶ Literature review
- ▶ Body of the paper
 - ▶ Theory (not discussed)
 - ▶ Empirical work
- ▶ Conclusion
- ▶ Appendices

Abstract

- ▶ Function of abstract: communicate the one central and novel contribution/finding
- ▶ 100-150 words limit - try to stick to it
- ▶ Say what you find not what you look for
 - ▶ “We investigate the effect of the XYZ regulation on house prices.”
 - ▶ “The XYZ regulation led to an increase in home-ownership as well as price.”
- ▶ Be specific and use numerics
 - ▶ “We find a significant and positive effect.”
 - ▶ “Using DiD for identification, the policy caused an increase in home-ownership by 5%.”

Introduction

- ▶ First sentence is hard (but important)
- ▶ Your paper has to be interesting on its own, not because others write a lot about XYZ
 - ▶ “The finance literature has been long interested in...”, And? Why?
- ▶ Explain contribution so that people understand it, do not just state it
 - ▶ “My results show that the XYZ-theory is rejected”, Why? What theory?
- ▶ Be specific on what you do and find

One potential structure of an introduction

- ▶ P1: Why is the RQ interesting?
- ▶ P2: Your main finding
- ▶ P3-5: How did you get there?
- ▶ P6-8: How it relates to other papers, how we can use your insight, is it relevant for public policy, . . . ?

Literature review

- ▶ Do not start introduction with it
- ▶ Two options
 - ▶ End of introduction after your findings
 - ▶ Own section
- ▶ Danger: It is hard enough to explain your own work in simple terms, do you want to explain what everyone else is doing in addition?
- ▶ However, credit where credit is due
- ▶ Function: set your paper off against the two/three closest current papers
 - ▶ Do not just list papers (history lesson)
 - ▶ Explain how your results improve/differ/build on the other paper (Be specific!)

Body of the paper

- ▶ Get to the central result as fast as possible
- ▶ Nothing before the main results that a reader does not need to know in order to understand the main result

Empirical work

- ▶ Introduce your data briefly
 - ▶ No excessive discussion of how control variables behave
 - ▶ But, if you have a new measure (e.g. Text based measure of ESG disclosure), you need to convince the reader that your measure of interest behaves as expected
- ▶ Start with the main result
- ▶ Do not motivate your specification by pointing out what did not work
- ▶ After the main result try to show mechanism, subsequently robustness
- ▶ Graphs are a good tool to show intuition

Conclusion

- ▶ In some sense not necessary, but you have to write it
- ▶ Results in the bigger scheme of things
 - ▶ What have we learned?
 - ▶ Who might have use for it?
 - ▶ Is it relevant for policy?
 - ▶ Implications of your findings
 - ▶ (maybe as a separate chapter)
- ▶ No plans for future research
- ▶ No meaningless shortcoming of your approach
 - ▶ Shortcomings that could have been addressed give a bad aftertaste
 - ▶ If shortcoming, tell what the consequence would be on your results, conclusions, etc.

Appendices

- ▶ They are great!
- ▶ Robustness, different empirical approaches, etc.
- ▶ A lot that is moved to the appendix, might be moved outside the paper eventually

Writing

General

- ▶ Shorter is better
 - ▶ Can I make the same point in less space?
 - ▶ Do not repeat things
- ▶ “First describe what you do, then explain it, compare it to alternatives, compare it to others’ procedures”
- ▶ Previews and recalls are a good sign of poor organization
- ▶ Strive for precision
- ▶ Document your work
 - ▶ Any reader should know how to reproduce your results
- ▶ Simple is better
 - ▶ Goal: reader needs to understand what you are doing
 - ▶ Do not dress up to look impressive

Tables

- ▶ Self-contained caption
 - ▶ Goal: Reader understands the result without looking at the text
 - ▶ Empirical model
 - ▶ Dependent variables
 - ▶ Less detail on independent variables
 - ▶ But, NOT substitute for detailed documentation in the body of the paper
- ▶ Revealing variables names, not what you use in your program
 - ▶ “Market Value” instead of “M.CAP.v3”
- ▶ Correct number of digits, not what the program spits out
- ▶ Use sensible units
 - ▶ 1.2 bp versus 0.00012

Figures

- ▶ “Make the paper alive”; great tool
- ▶ Self-contained caption
- ▶ Label axis
- ▶ Sensible units

Writing tips

- ▶ Keep track of what the readers knows and does not know
- ▶ Expect a certain level of econometric understanding
 - ▶ E.g. Do not explain OLS, but maybe the effect of Industry Fixed effects on interpretation
- ▶ Active tense and concise
 - ▶ “It was assumed. . .” Who?
 - ▶ “Data was constructed. . .” By whom?
 - ▶ “In the table the coefficient can be seen..” By whom?
 - ▶ “The results might be interpreted. . .” Or not?
- ▶ Bad writing often is caused by trying to avoid responsibility for what one is saying

Writing tips

- ▶ Present tense works well referring to your own work in the paper as well as previous work
- ▶ Avoid technical jargon if possible
- ▶ Be concrete
 - ▶ Maybe a concrete example works
- ▶ Avoid “think” and “feel” when presenting arguments

Advice for empirical work

Advice for empirical work

- ▶ How to do empirical work?
- ▶ Narrow and interesting research question
- ▶ The most important part is identification
 - ▶ Usually A causes/affects/correlates with B
 - ▶ Explain how you are able to identify the relationship
- ▶ Economic mechanism that causes variation in your right hand variable
 - ▶ Does this correspond to your specification?

Advice for empirical work

- ▶ Consider which controls to be included in the regression
 - ▶ “Throw baby out with the bathwater”
 - ▶ Example: Effects of education on wage controlling for industry and/or job level
- ▶ Discuss the usual suspects w.r.t. potential violations of the zero conditional mean assumption
 - ▶ Simultaneity
 - ▶ Reverse causality
 - ▶ Omitted variable
 - ▶ What would have been the ideal experiment. How does reality differ from this? Is there a problem?
 - ▶ Be specific: What variable? What is the effect?
- ▶ Explain economic significance

Summary

- ▶ Grain of salt: taste, preferences, and experience
- ▶ Main take-away: Make it easy for the reader to understand exactly what you are doing
- ▶ False premise: We do research/exams/assignments/etc., and then just write up our results.

Material

This lecture is build around “Writing Tips for Ph.D. students” by John H. Cochrane
([link](#))